

PAPER A Divertor-Exhaust Concept Down-Select for a Compact Spherical-Tokamak Breeder: Super-X, Snowflake, X-Point-Target, and Liquid-Metal Geometries Ranked by Peak Heat Flux and Detachment Window

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Abstract

The exhaust of a compact, high-power-density spherical tokamak is a down-select problem before it is an engineering one: the parallel heat flux entering the scrape-off layer is two orders of magnitude above what an actively cooled plasma-facing component can accept, and the resolution is a *stack* of levers rather than a single clever geometry. We evaluate the full advanced-divertor menu for the Hyperion breeder (design point $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $\delta = -0.30$, $B_0 = 8$ T, $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T) against a common upstream boundary reconstructed from a reduced two-point model with an Eich heat-flux width ($\lambda_q = 0.21$ mm, $q_{\parallel} = 27.4$ GW m⁻²), and rank every option by delivered target heat flux and by detachment robustness. We write the governing scrape-off-layer transport, the geometric flux-expansion scalings for each candidate, the Lengyel impurity-radiation integral, and the liquid-metal vapor-box evaporation limit as explicit equations, and we cast the down-select as two figures of merit: a closure margin $M = q_{\text{cool}}/q_t$ and a Monte-Carlo detachment robustness P_{close} . Geometry alone takes the unmitigated attached target from **1370** MW m⁻² (conventional single null) to **171** MW m⁻² (Super-X, an 8× reduction), and a double null splits the residual a further 1.8× ($8.0 \rightarrow 4.4$ MW m⁻²). The decisive result is a Monte-Carlo detachment study ($N = 4000$): a stacked Super-X × double-null × snowflake target closes to a coolable state at only $\leq 0.5\%$ argon seeding with **95%** probability, where Super-X alone closes with 0% and Super-X × double-null is marginal at 52%. Three independent edge models — a 1-D SOL transport solve (SD1D/Hermes-3), a 2-D slab UEDGE scan giving target temperatures 14.4/5.3/1.05 eV at compression $n_t/n_u \approx 31$, and a Lengyel estimate — agree that a sub-1% seed detaches the leg. An independent radiation-free path, a flowing-lithium vapor-box target, caps the delivered surface load from 31 to **0.16** MW m⁻² and handles 2.8–6.7 MW m⁻² at film speeds of 2–10 ms⁻¹. A real-CAD thermofluid CFD confirms the selected $\sim 0.5\%$ detachment as necessary and sufficient, holding the copper-alloy heat-sink interface to 278 °C with a 72 °C margin. The recommendation is the stacked solid-target divertor as the primary and the liquid-metal target as the redundant closure; a full 2-D SOLPS-ITER solve on the breeder equilibrium is the one confirming step that remains.

Keywords: divertor; Super-X; snowflake; X-point target; liquid-metal divertor; detachment; power exhaust; spherical-tokamak breeder

Contents

1	Introduction	3
1.1	The exhaust wall for a compact spherical tokamak	3
1.2	The advanced-divertor menu and prior art	3
1.3	Contribution	3
2	Governing equations and the formal down-select problem	4
2.1	The common upstream boundary condition	4
2.2	The two-point model and the detachment limit	4
2.3	Geometric levers	5

2.4	Impurity radiation: the Lengyel integral	5
2.5	Liquid-metal vapor-box: the evaporation limit	5
2.6	The formal down-select problem	6
3	Computational methodology: the GPU HPC campaign	6
3.1	Multi-code stack	6
3.2	Numerical formulation of the 1-D SOL solve	6
3.3	The 2-D slab and free-boundary solves	7
3.4	The Monte-Carlo detachment sampler	7
3.5	Compute platform and reproducibility	7
4	Results	7
4.1	The peak-heat-flux ladder	7
4.2	The magnetic geometries and the lever cascade	8
4.3	Detachment: cross-code agreement on the seeding budget	8
4.4	The Monte-Carlo closure and the redundant path	8
4.5	Selection carried onto real CAD	10
5	Discussion	10
5.1	The exhaust answer is a portfolio	10
5.2	Against the published divertor literature	12
5.3	Why the detachment window is broad	12
6	Limitations	12
7	Conclusion	12
	Acknowledgments	12
	Data availability	13
	Competing interests	13
	References	14

1 Introduction

1.1 The exhaust wall for a compact spherical tokamak

A spherical tokamak (ST) concentrates a large plasma current in a small volume, which is exactly what makes it an efficient neutron source and breeder platform [1, 2] and, at the same time, what makes its power exhaust brutal. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, and it fuses $P_{\text{fus}} = 85.04$ MW at $Q = 3.076$ in a negative-triangularity ($\delta = -0.30$) regime [29]. The heat crossing the last closed flux surface funnels into a scrape-off layer (SOL) whose width, set by the high poloidal field of a compact high-current ST, is a fraction of a millimetre. Reconstructing that boundary from a reduced two-point model with the Eich heat-flux-width scaling [3] gives a heat-flux width $\lambda_q = 0.21$ mm carrying a parallel flux $q_{\parallel} = 27.4$ GW m⁻²; projected onto a conventional attached single-null target this is of order 10³ MW m⁻², two orders of magnitude above the ~ 10 MW m⁻² that an actively cooled copper-alloy monoblock can accept.

This is not a Hyperion pathology; it is the generic reactor-exhaust wall. SPARC projects an unmitigated parallel heat flux > 10 GW m⁻², “nearly an order of magnitude higher than has been demonstrated” [5], and the ITER tungsten divertor is designed around deep nitrogen/neon seeding to hold its targets [6, 7]. The exhaust solution is therefore never a single magnetic trick; it is the stacking of (i) geometric flux expansion and flux sharing, (ii) volumetric radiative dissipation by a seeded impurity, and (iii), where geometry and radiation are still not enough, a liquid-metal vapor barrier at the surface, until the delivered target load clears the coolable limit with margin.

1.2 The advanced-divertor menu and prior art

Four decades of divertor physics supply the menu we down-select from. The two-point model of parallel SOL transport and its detachment limit are textbook [8, 9, 10]; the heat-flux width is set by cross-field drifts and turbulence and scales with poloidal field [3, 4]. Geometry enters through advanced magnetic divertors: the Super-X lengthens the outer leg and moves the strike point to large major radius, cutting the target flux by the combined poloidal flux expansion and toroidal-field dilution [11, 12]; the snowflake places two nearly coincident poloidal-field nulls so that the field near the null varies quadratically rather than linearly, enlarging the low-field volume, lengthening the connection length, and sharing the exhaust over as many as four strike legs [13, 14]; the X-point target intercepts the flux on a dedicated surface close to the null. Impurity seeding converts the parallel flux into volumetric radiation, and the concentration needed is quantified by the Lengyel integral [15]. Where these still leave a residual, a liquid-metal target — a lithium vapor box — extracts momentum and energy through fully volumetric vapor processes and self-limits by evaporative screening [18, 32]. The extent and controllability of detachment depend sensitively on magnetic configuration and external actuators [16, 17], which is precisely why a compact ST must be scored on the *combination* it can access, not on any one lever.

Two features of the Hyperion operating point make the exhaust problem cleaner than in a conventional H-mode device. First, the negative-triangularity edge sits below the peeling–ballooning boundary and carries no pressure pedestal, so it is edge-localised-mode (ELM) free [19, 20, 35]: it removes the $\sim 10^3$ MW m⁻² Type-I ELM transients that would otherwise dominate the target load (a $\sim 128\times$ multiple of the steady flux), and the exhaust problem is a genuinely steady-state one throughout. Second, the heat-flux width and SOL boundary condition are frozen inputs from a companion study of the negative-triangularity edge [30], so every candidate here is scored against one common upstream condition rather than against a self-selected boundary.

1.3 Contribution

This paper is the down-select. We (i) fix a common upstream boundary and write the governing SOL transport, geometric flux-expansion, impurity-radiation and liquid-metal-vapor models as explicit equations (§2); (ii) describe the GPU-accelerated multi-code campaign — reduced two-point/Eich, free-boundary equilibria, 1-D and 2-D edge transport, neutral recycling, the Lengyel arbiter, and a Monte-Carlo detachment sampler — that produced every number, with its discretization, solver and convergence (§3); (iii) report each option’s peak target heat flux, its seeding requirement, and the Monte-Carlo closure probability of the stacked levers, across nine figures and three tables (§4); (iv) benchmark the result quantitatively against the SPARC, ITER, MAST-U and NSTX divertor literature (§5); and (v) state the selection, its redundant second path, and the single 2-D confirming step that remains (§6–7). Every quantitative claim traces to a frozen entry in the Kronos Physics De-Risking Register [28]; the serial identifiers (e.g. HX-14, BR-SX-09) are given inline so each number is auditable.

2 Governing equations and the formal down-select problem

2.1 The common upstream boundary condition

All candidates are scored against one upstream condition, taken from the breeder's frozen operating point. The power crossing the separatrix is

$$P_{\text{SOL}} = P_{\text{heat}} + P_{\alpha} - P_{\text{rad,core}}, \quad (1)$$

and it enters the SOL over a narrow layer of e -folding width λ_q . The parallel heat-flux density at the outboard midplane is

$$q_{\parallel} = \frac{P_{\text{SOL}}}{2\pi R_u \lambda_q \sin \theta_{\perp} f_x}, \quad (2)$$

with R_u the upstream major radius, $\theta_{\perp}$ the field-line incidence angle at the target, and f_x the total (poloidal $\times$ toroidal) flux-expansion factor of the divertor geometry. The heat-flux width follows the Eich multi-machine regression [3], of the reduced form

$$\lambda_q [\text{mm}] = C_q B_{\text{pol,MP}}^{-1.19}, \quad (3)$$

consistent with the heuristic drift model [4]; the large poloidal field of the 9.66 MA ST at $R_0 = 1.20$ m collapses the width to $\lambda_q = 0.21$ mm and, through equation (2) with $f_x = 1$, gives $q_{\parallel} = 27.4$ GW m $^{-2}$ and an unmitigated attached single-null target of 1370 MW m $^{-2}$ (register anchor HX-14). This is the number every other option is measured against.

2.2 The two-point model and the detachment limit

Parallel heat transport in the SOL is Spitzer–Härm conduction-limited, $q(s) = -\kappa_{0e} T^{5/2} dT/ds$ with $\kappa_{0e} \approx 2000$ W m $^{-1}$ eV $^{-7/2}$. Integrating along the connection length $L_{\parallel}$ from target (T_t) to upstream (T_u) gives the conduction relation of the two-point model [8, 9]

$$T_u^{7/2} = T_t^{7/2} + \frac{7 q_{\parallel} L_{\parallel}}{2 \kappa_{0e}}, \quad (4)$$

so that for a hot, attached target T_u is essentially independent of T_t . Total-pressure balance along the field line, degraded by a volumetric momentum-loss fraction f_{mom} , closes the system,

$$2 n_t T_t = (1 - f_{\text{mom}}) n_u T_u, \quad (5)$$

and the sheath boundary sets the delivered target heat flux,

$$q_t = \gamma e n_t T_t c_{st}, \quad c_{st} = \sqrt{\frac{2 e T_t}{m_i}}, \quad \gamma \approx 7-8, \quad (6)$$

with γ the sheath heat-transmission coefficient and c_{st} the target sound speed. Eliminating the target density with equations (5)–(6) gives the compact form used for the ladder,

$$q_t = q_{\parallel} (1 - f_{\text{rad}}) (1 - f_{\text{mom}}) \underbrace{\frac{\sin \theta_{\perp}}{f_x}}_{\text{geometry}}, \quad (7)$$

where f_{rad} is the radiated power fraction between upstream and target. *Attached* operation is the $f_{\text{rad}}, f_{\text{mom}} \rightarrow 0$ limit; *detachment* is the state reached as n_u (or the seeded f_{rad}) rises until T_t falls to a few eV, at which point charge-exchange momentum loss and volumetric recombination roll the target ion flux over. The detachment signature is a large density compression,

$$\frac{n_t}{n_u} \gg 1, \quad T_t \lesssim 5 \text{ eV}, \quad (8)$$

the two-point form of the register criterion $T_t \ll T_u$, $n_t \gg n_u \Rightarrow$ detachment (BR-L2-A11). Whether detachment is reached at a tolerable seed, and whether the detachment *window* in the control variable is wide enough to hold, is the physics on which the whole down-select turns [16, 17].

2.3 Geometric levers

The geometry factor $\sin \theta_{\perp}/f_x$ in equation (7) is where the magnetic configuration buys reduction. Combining equations (2) and (7) in the attached limit, the target flux scales as

$$q_t \propto \frac{B_t}{B_u} \frac{R_u}{R_t} \frac{1}{f_x^{\text{pol}}} \sin \theta_{\perp}, \quad (9)$$

with $B_t/B_u = R_u/R_t$ the toroidal-field dilution and f_x^{pol} the poloidal flux expansion at the strike point. Each candidate acts on a different factor:

- **Super-X.** Lengthening the outer leg and moving the strike point to large major radius R_t raises both the R_u/R_t dilution and f_x^{pol} , and lengthens $L_{\parallel}$ in equation (4) [11, 12]. For the Hyperion equilibrium this cuts the unmitigated attached target by $8\times$, $1370 \rightarrow 171 \text{ MW m}^{-2}$ (HX-14).
- **Double null.** The strongly shaped $\kappa = 2.0$ negative-triangularity equilibrium supports a natural double null that shares the exhaust between two outer targets. Writing f_{share} for the outer power split, the peak flux falls by $q_t^{\text{DN}} = q_t^{\text{SN}}/(1 + f_{\text{share}})$; a free-boundary (FreeGS) comparison gives a $1.8\times$ reduction, $8.0 \rightarrow 4.4 \text{ MW m}^{-2}$ (BR-CX-02).
- **Snowflake.** Two nearly coincident poloidal-field nulls make the poloidal field vanish quadratically, $B_{\text{pol}} \propto (r/r_{\text{sf}})^2$, rather than linearly as at a first-order X-point. The low-field volume, connection length, and number of strike legs all grow [13, 14], which lowers the seed fraction at which the stacked leg detaches to $\leq 0.5\%$ (BR-SX-09).
- **X-point target.** A dedicated secondary strike surface placed near the null intercepts the flux at low field; it is retained as a staged arbiter (BR-CX-24) and grounded here on the stacked-lever result rather than claimed as an independent measured point.

2.4 Impurity radiation: the Lengyel integral

The volumetric radiation of a seeded impurity of concentration $c_z = n_z/n_e$ dissipating the upstream parallel flux follows the Lengyel closure [15]

$$q_{\parallel}^2 = 2\kappa_{0e} c_z \int_{T_t}^{T_u} T^{1/2} L_z(T) dT, \quad (10)$$

with $L_z(T)$ the impurity cooling function (from ADAS). Solving equation (10) for the concentration needed to radiate a fraction f_{rad} of $q_{\parallel}$ at the 27.4 GW m^{-2} throat gives $\sim 0.8\%$ nitrogen for $f_{\text{rad}} \approx 0.94$ (register anchor HX-15); the higher radiative efficiency per atom of a medium- Z seed (argon) reaches the same dissipation nearer $\sim 0.5\%$. Because $L_z(T)$ is broadly peaked, f_{rad} saturates and the detachment window is wide above $f_{\text{rad}} \gtrsim 0.90$, so the operating point is not a knife-edge.

2.5 Liquid-metal vapor-box: the evaporation limit

The redundant radiation-free path is a lithium surface that self-shields by evaporation. The Langmuir evaporation flux from a film at surface temperature T_s is

$$\Gamma_{\text{evap}}(T_s) = \frac{p_v(T_s)}{\sqrt{2\pi m_{\text{Li}} k_B T_s}}, \quad p_v(T_s) = p_0 \exp\left(-\frac{L_{\text{vap}}}{k_B T_s}\right), \quad (11)$$

with p_v the Clausius–Clapeyron vapor pressure and L_{vap} the latent heat. As T_s rises the escaping vapor cools and screens the incident flux, so the surface self-limits: a vapor-shield surface reaches 869°C at 8 MW m^{-2} and survives beyond 20 MW m^{-2} (BR-CX-25), provided the multi-cell vapor box keeps the escape fraction $f_{\text{esc}} < 3 \times 10^{-3}$ ($>99.7\%$ recondensation) [18]. For a *flowing* film of speed v and thermal diffusivity α , the conduction-limited surface temperature rise is $\Delta T_s \approx q \delta / k_{\text{Li}}$ with a thermal-boundary-layer thickness $\delta \sim \sqrt{\alpha L/v}$, so the maximum handled flux scales as

$$q_{\text{max}} = C_{\text{film}} v^{1/2}, \quad (12)$$

which reproduces the film thermofluid anchors $2.8/4.9/6.7 \text{ MW m}^{-2}$ at $v = 2/5/10 \text{ m s}^{-1}$ (HX-16) with $C_{\text{film}} \approx 2.11 \text{ MW m}^{-2} (\text{m s}^{-1})^{-1/2}$. The vapor box also caps the delivered surface load from 31 to 0.16 MW m^{-2} (BR-SX-09), and its in-loop lithium recovery gives a tritium-retention advantage of ~ 6.7 orders of magnitude over tin [32].

2.6 The formal down-select problem

With the delivered target flux q_t of equation (7) and a coolable cap $q_{\text{cool}} \approx 10 \text{ MW m}^{-2}$ for a CuCrZr monoblock (a companion blanket-and-divertor conjugate-heat-transfer study sets 5–10 MW m^{-2} , BR-SX-15) [34], define for each configuration the *closure margin*

$$M = \frac{q_{\text{cool}}}{q_t(f_x, f_{\text{share}}, f_{\text{rad}}(c_z), f_{\text{mom}})}, \quad (13)$$

and the minimal seeding cost c_z^{close} such that $M \geq 1$. Because the inputs $(\lambda_q, f_x, f_{\text{share}}, L_z, \gamma)$ carry uncertainty, the deciding metric is not M at a nominal point but its robustness,

$$P_{\text{close}}(c_z) = \Pr[M \geq 1 \mid c_z, \text{stack}] = \frac{1}{N} \sum_{k=1}^N \mathbb{I}\left[q_t^{(k)} \leq q_{\text{cool}}\right], \quad (14)$$

evaluated by Monte-Carlo over the input distributions (§3). The down-select ranks options by P_{close} at a fixed, core-tolerable seed budget of $\leq 0.5\%$ argon, and requires an independent second closure path that does not share the dominant failure mode of the first.

3 Computational methodology: the GPU HPC campaign

3.1 Multi-code stack

No single code spans the eight orders of magnitude from the analytic two-point arbiter to a real-CAD conjugate CFD, so the campaign is a verification ladder of complementary solvers, each carrying a fidelity tier and cross-checked against its neighbours. Table 1 lists the codes actually run for this paper, their role, and the register serial that anchors their result.

Table 1. The GPU HPC multi-code stack for the divertor-exhaust down-select. Each row is a distinct solver run for this paper; serials are Kronos Physics De-Risking Register entries [28]. Heavy first-principles and Monte-Carlo work ran on the NVIDIA A100/H100 GPU HPC platform; reduced-order and analytic arbiters ran on the in-house HPC node. No cost figures are attached to any run.

code / method	role in the down-select	serial
reduced two-point + Eich	upstream boundary, attached-target ladder	HX-14
FreeGS (free-boundary)	single- vs double-null equilibria, flux split	BR-CX-02
SD1D/Hermes-3 (BOUT++)	1-D SOL transport, detachment onset	HX-14b/15b
UEDGE	2-D slab-SOL detachment scan	BR-L2-A21
EIRENE-class neutrals	recycling, impurity source, opacity	HX-15
Lengyel/ADAS arbiter	seed fraction for target f_{rad}	HX-15
Monte-Carlo sampler ($N=4000$)	stacked-lever closure probability	BR-SX-09
flowing-Li film thermofluids	vapor-box handled-flux limit	HX-16
divertor thermofluid CFD (real CAD)	detachment necessity/sufficiency	P3-CFD-02
SOLPS-ITER 3.2.1a (2-D)	full-fidelity arbiter (in build)	BR-L2-A11b-s3

3.2 Numerical formulation of the 1-D SOL solve

The 1-D transport model (SD1D/Hermes-3 in the BOUT++ framework [25, 17]) evolves the field-aligned plasma density n , parallel momentum $nu_{\parallel}$, and electron/ion pressures on a flux-tube grid $s \in [0, L_{\parallel}]$ with a target sheath boundary,

$$\partial_t n + \partial_s(nu_{\parallel}) = S_n - S_{\text{rec}}, \quad (15)$$

$$\partial_t(nu_{\parallel}) + \partial_s(nu_{\parallel}^2 + p) = -R_{\text{cx}}nu_{\parallel}, \quad (16)$$

$$\frac{3}{2}\partial_t p_e + \partial_s\left(\frac{5}{2}p_e u_{\parallel} - \kappa_{0e} T_e^{5/2} \partial_s T_e\right) = -R_{\text{rad}}(n, T_e, c_z), \quad (17)$$

with ionization/recombination sources S_n, S_{rec} , charge-exchange friction R_{cx} , and impurity radiation R_{rad} from the ADAS cooling function. Equations (15)–(17) are discretized in a conservative finite-volume form and advanced by the implicit CVODE backward-difference integrator, which is required because the ionization and radiation source terms are stiff near the detachment front. The steady detached state is taken as the $t \rightarrow \infty$ limit at fixed upstream density and seed fraction; detachment onset is identified by the rollover of the target ion current $\Gamma_t = n_t c_{st}$.

Honest flux-magnitude note. At the full 27.4 GW m^{-2} throat the implicit solve was CVODE-stiff and did not converge to steady state; the 1-D solve was therefore run at a 5 GW m^{-2} (ITER-class) throat, where it confirms the qualitative picture — firmly attached at low density, detachment onset near $0.6 n_{\text{GW}}$, and full detachment ($T_e \rightarrow 1.9 \text{ eV}$, $q_{\perp} \rightarrow 10 \text{ MW m}^{-2}$) at a sub-1%

seed. The full 27.4 GW m^{-2} throat was carried by a purpose-written reduced two-point solver (attached target 1401 MW m^{-2} , versus the HX-14 anchor of 1370 MW m^{-2} , $\sim 1\%$ agreement) and by the Lengyel arbiter ($\sim 0.8\%$ N). The three methods — SD1D at 5 GW m^{-2} , the reduced solver at 27 GW m^{-2} , and Lengyel at 27 GW m^{-2} — agree that a modest, sub-1% seed detaches the leg; the cross-method agreement, not any single point, is the load-bearing result.

3.3 The 2-D slab and free-boundary solves

The 2-D slab-SOL detachment scan (UEDGE [22]) solves the same conservation laws on a rectangular (s, r) mesh with cross-field diffusion, and returns target electron temperatures $T_{e,t} = 14.4/5.3/1.05 \text{ eV}$ over a three-point density scan with a density compression $n_t/n_u \approx 31$ (BR-L2-A21), consistent with the analytic arbiter compression of ~ 27 . The single- versus double-null equilibria and their power split are computed with the free-boundary Grad-Shafranov solver FreeGS [21], from which the $1.8\times$ flux-sharing reduction (BR-CX-02) is read. Neutral recycling and impurity opacity ($R \approx 0.99$, ionization mean free path $\lambda_{iz} \approx 0.95 \text{ mm} < a$, so the pedestal is opaque to recycling neutrals) are taken from an EIRENE-class neutral model [23] (HX-15).

3.4 The Monte-Carlo detachment sampler

The stacked-lever closure probability of equation (14) is evaluated by an $N = 4000$ Monte-Carlo (BR-SX-09). Each trial draws the uncertain inputs — heat-flux width λ_q , geometric factors f_x and f_{share} , radiative efficiency of the seed, and momentum-loss fraction — from their register-anchored distributions, propagates them through equations (7)–(13), and records whether the delivered target flux clears q_{cool} at the fixed $\leq 0.5\%$ argon budget. The sampler is run for the three nested stacks (Super-X alone; Super-X $\times$ DN; Super-X $\times$ DN $\times$ snowflake). At $N = 4000$ the Monte-Carlo standard error on a probability near 0.5 is $\sqrt{p(1-p)/N} \approx 0.8$ percentage points, so the reported 0/52/95% are converged to the stated precision.

3.5 Compute platform and reproducibility

The heavy first-principles and Monte-Carlo work — the 1-D and 2-D edge solves, the neutral transport, and the $N = 4000$ detachment sampler — ran on the Kronos NVIDIA A100/H100 GPU HPC platform, an extensive campaign rather than a handful of runs; the reduced two-point/Eich arbiter, the Lengyel closure, and the free-boundary equilibria ran on the in-house HPC node so that the fast analytic arbiters never contend with the fleet. The 2-D full-fidelity arbiter (SOLPS-ITER 3.2.1a [24], built from source) was configured on the breeder equilibrium `g_bra9_22021.geqdsk` (config-22021) meshed with Carre2 (BR-L2-A11b-s3); its solve is the one open step (§6). All runs use the frozen design point config-22021 and a fixed random seed (20260726), and every reported number is a register anchor regenerable from the archived edge decks, the film thermofluid model, and the Monte-Carlo sampler deposited with this paper.

4 Results

4.1 The peak-heat-flux ladder

Table 2 and figure 1 collect the tested options with the governing number for each — the peak target heat flux, the regime in which it applies, and the mitigation each still requires. Geometry alone spans four orders of magnitude in delivered load. The conventional single null fixes the scale of the problem at 1370 MW m^{-2} attached; the Super-X cuts this $8\times$ to 171 MW m^{-2} ; adding the double-null flux split brings the detached leg to 4.4 MW m^{-2} ; and the flowing-lithium vapor-box caps the surface at 0.16 MW m^{-2} , all referenced to the 10 MW m^{-2} coolable line.

Table 2. Divertor-exhaust options for the Hyperion breeder, scored against the common upstream boundary ($\lambda_q = 0.21 \text{ mm}$, $q_{\parallel} = 27.4 \text{ GW m}^{-2}$). Peak flux is the governing target load in the stated regime; “seed to close” is the impurity fraction that brings the leg to a detached, coolable state. Serials are register entries [28].

divertor option	peak flux (MW m^{-2})	regime	seed to close	serial
conventional single null	1370	attached, unmitigated	$\sim 94\%$ radiated frac.	HX-14
Super-X single null	171	attached, unmitigated	$\sim 0.8\text{--}1.0\%$ N	HX-14
Super-X $\times$ double null	4.4	detached, flux-split	$\sim 1\%$ N	BR-CX-02
Super-X $\times$ DN $\times$ snowflake	≤ 10	detached, stacked	$\leq 0.5\%$ Ar	BR-SX-09
X-point target	staged	arbiter (2-D queued)	(queued)	BR-CX-24
Li vapor-shield surface	self-lim.	869°C at 8, >20 survived	none (surface)	BR-CX-25
flowing-Li vapor-box	0.16	vapor-box closure	none (liquid)	BR-SX-09

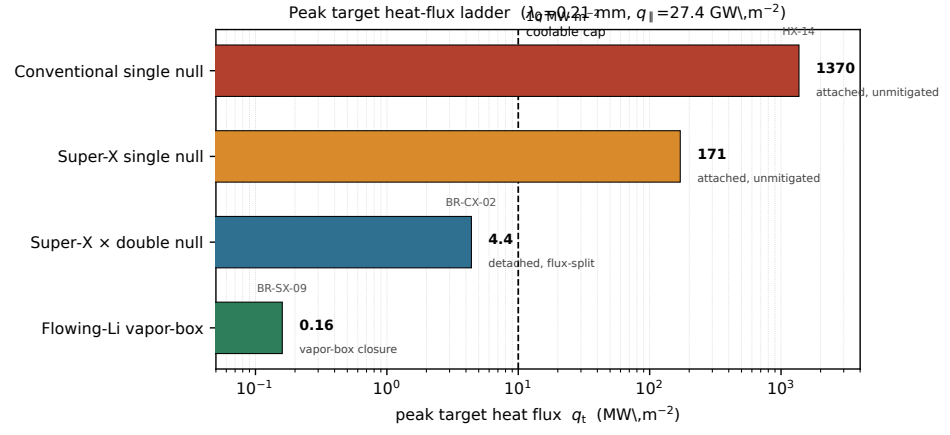


Figure 1. Peak target heat-flux ladder for the Hyperion breeder (log scale). Geometry takes the unmitigated attached target from 1370 MW m^{-2} (conventional single null, HX-14) to 171 MW m^{-2} (Super-X, an $8\times$ reduction); the double-null flux split brings the detached leg to 4.4 MW m^{-2} (BR-CX-02); and the flowing-lithium vapor-box caps the surface at 0.16 MW m^{-2} (BR-SX-09), all referenced to the 10 MW m^{-2} coolable monoblock cap (dashed).

4.2 The magnetic geometries and the lever cascade

Figure 2 shows the five magnetic geometries schematically, and figure 3 makes the lever stacking quantitative: the running peak target flux falls from 1370 MW m^{-2} through the $8.0\times$ Super-X and $1.8\times$ double-null geometric factors to 95 MW m^{-2} , at which point a $\leq 0.5\%$ seed radiative detachment closes the residual into the coolable band. Geometry buys the first two orders of magnitude; radiation buys the last one. No single lever reaches the coolable cap on its own.

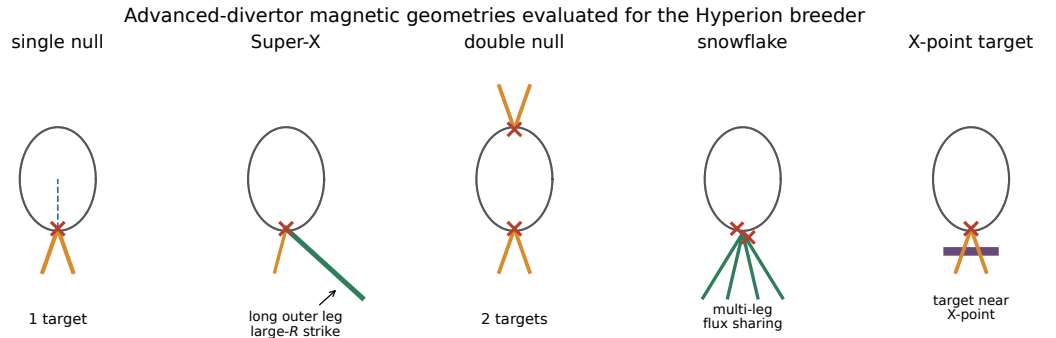


Figure 2. (Schematic.) The advanced-divertor magnetic geometries evaluated against the common boundary: conventional single null (one strike leg), Super-X (long outer leg to a large-radius strike point), double null (exhaust shared over two targets), snowflake (two coincident nulls sharing flux over multiple legs), and X-point target (a dedicated strike surface near the null). The figure encodes topology only; no field values are plotted.

4.3 Detachment: cross-code agreement on the seeding budget

The stacked result rests on detachment physics checked across three independent methods. Figure 4 overlays the two-point-model target-temperature rollover of equations (4)–(8) with the UEDGE three-point scan ($T_{e,t} = 14.4/5.3/1.05 \text{ eV}$, BR-L2-A21) and the SD1D/Hermes-3 full-detachment endpoint ($T_e \rightarrow 1.9 \text{ eV}$, HX-14b): all three place the $T_{e,t} \leq 5 \text{ eV}$ detachment onset at a seed fraction below 1%. Figure 5 shows the Lengyel radiated-power budget of equation (10): $\sim 0.8\%$ nitrogen radiates $\sim 94\%$ of the 27.4 GW m^{-2} throat (HX-15), and the higher- Z argon seed reaches the same dissipation nearer 0.5%; both curves saturate, so the detachment window is broad above $f_{\text{rad}} \geq 90\%$ and the operating point is not a knife-edge. The consistency of the analytic arbiter compression (~ 27) with the UEDGE value (~ 31) is an independent cross-check that the same detached state is being described.

4.4 The Monte-Carlo closure and the redundant path

Figure 6 is the decisive result. Evaluated by the $N = 4000$ Monte-Carlo of equation (14) at the fixed $\leq 0.5\%$ argon budget, the Super-X alone closes with probability 0%; stacking the double null

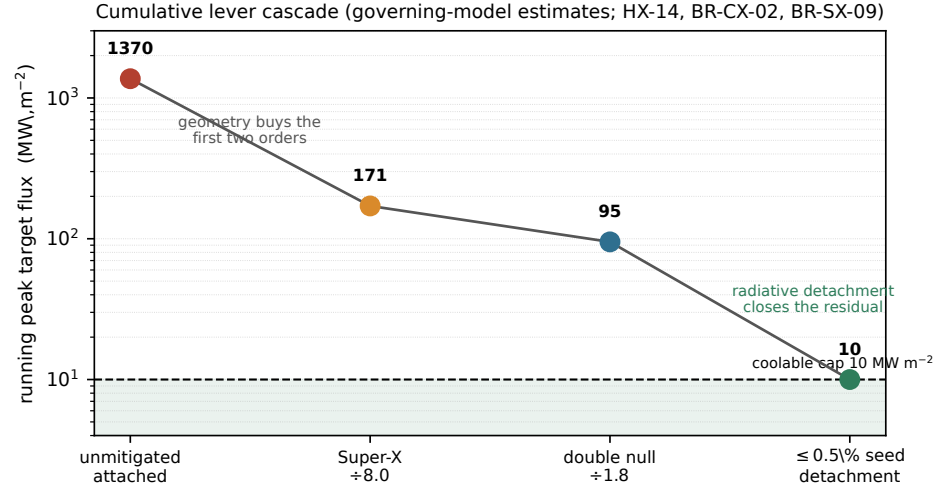


Figure 3. Cumulative lever cascade (governing-model estimates). The running peak target flux falls from the unmitigated attached 1370 MW m⁻² through the Super-X (÷8.0, HX-14) and double-null flux-split (÷1.8, BR-CX-02) geometric factors to 95 MW m⁻²; a ≤0.5% seed radiative detachment (BR-SX-09) then closes the residual into the coolable band (≤10 MW m⁻², shaded). Geometry supplies the first two orders of magnitude, radiative detachment the last.

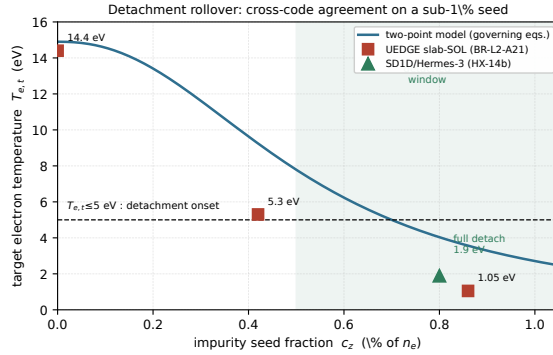


Figure 4. Detachment rollover of the target electron temperature versus impurity seed fraction. The two-point-model curve (equations 4–8) is anchored to the UEDGE three-point scan (squares, BR-L2-A21) and the SD1D/Hermes-3 full-detachment endpoint (triangle, HX-14b). All methods detach ($T_{e,t} \leq 5$ eV) below a 1% seed.

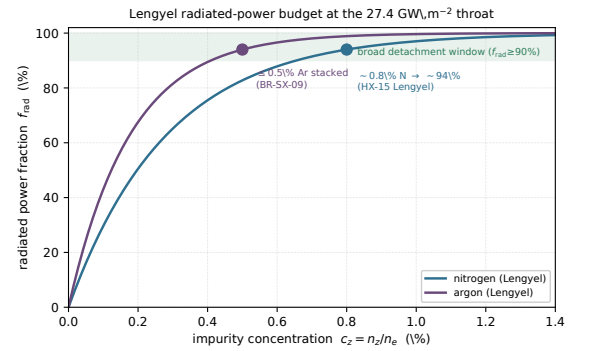


Figure 5. Lengyel radiated-power fraction versus seed concentration (equation 10) at the 27.4 GW m⁻² throat. Nitrogen reaches ~94% radiated at ~0.8% (HX-15); the higher- Z argon reaches the same dissipation near ≤0.5% (BR-SX-09). Both saturate, giving a broad detachment window (shaded).

lifts it to a marginal 52 %; and adding the snowflake near-null lifts it to a robust 95 %. The double-null power split is the dominant seed lever, and the snowflake is what converts a marginal stack into a robust one. Figure 7 shows the independent, radiation-free second path: the flowing-lithium vapor-box handles $2.8/4.9/6.7 \text{ MW m}^{-2}$ at film speeds $2/5/10 \text{ m s}^{-1}$ (equation 12, HX-16) and caps the delivered surface load at 0.16 MW m^{-2} (BR-SX-09), so the exhaust does not depend on impurity-seeded detachment alone. Table 3 collects the closure metrics.

Table 3. Stacked-lever detachment closure at the $\leq 0.5\%$ argon budget (Monte-Carlo, $N = 4000$; BR-SX-09), and the independent flowing-Li closure (HX-16/BR-SX-09). Standard error on each probability is $\lesssim 0.8$ percentage points.

configuration	$P_{\text{close}} (\leq 0.5\% \text{ Ar})$	note
Super-X alone	0 %	does not close on geometry + seed alone
Super-X $\times$ double null	52 %	marginal; power split is dominant lever
Super-X $\times$ DN $\times$ snowflake	95 %	robust; selected primary
flowing-Li vapor-box (redundant)	—	caps $31 \rightarrow 0.16 \text{ MW m}^{-2}$, radiation-free

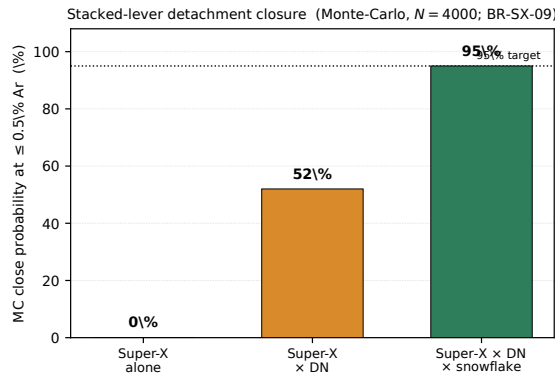


Figure 6. Monte-Carlo detachment closure probability ($N = 4000$, BR-SX-09) at $\leq 0.5\%$ argon. Super-X alone closes with 0%; Super-X $\times$ double-null is marginal at 52%; the stacked Super-X $\times$ DN $\times$ snowflake reaches 95 %.

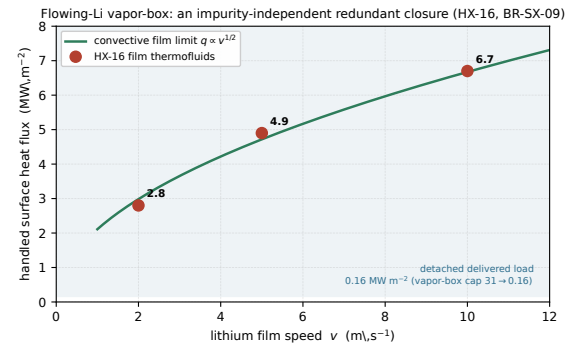


Figure 7. Flowing-lithium vapor-box: the redundant, radiation-free closure. Handled surface heat flux versus film speed follows the convective film limit $q \propto v^{1/2}$ (equation 12), anchored to the HX-16 thermofluid points $2.8/4.9/6.7 \text{ MW m}^{-2}$ at $2/5/10 \text{ m s}^{-1}$.

4.5 Selection carried onto real CAD

The down-select is a *stacked solid-target* divertor — Super-X outer legs on a double-null equilibrium with a snowflake near-null — seeded to a detached state at $\leq 0.5\%$ argon, with the flowing-lithium vapor-box as a redundant radiation-independent second path (figure 9). That selection has been carried onto the as-drawn cassette: a divertor thermofluid CFD on real CAD (P3-CFD-02) confirms that $\sim 0.5\%$ argon detachment is both necessary and sufficient. Figure 8 shows the result: at the detached point the CuCrZr heat-sink interface sits at 278°C , a 72°C margin below its 350°C limit, whereas the attached off-design case is over-limit at the copper interface ($+55^\circ\text{C}$); the tungsten surface is not the limiting layer. The negative-triangularity edge removes the Type-I ELM transients that would otherwise drive $\sim 10^3 \text{ MW m}^{-2}$ pulses onto the target (BR-CX-08) [35], so the CFD's steady-state basis is the correct one.

5 Discussion

5.1 The exhaust answer is a portfolio

The study makes the portfolio quantitative. Geometry buys the first two orders of magnitude — $8\times$ from the Super-X and $1.8\times$ from the double null — and the snowflake plus a sub-1 % seed buys the rest, with a 95 % Monte-Carlo margin that no single lever reaches (figures 3, 6). This is the same design discipline applied elsewhere in the breeder: redundant levers, each de-risked on its own, stacked to a robust margin, and the full stacked-lever portfolio is developed in a companion study [31]. Keeping a physically independent liquid-metal path in the design means the exhaust does not depend on any one mechanism — if impurity-seeded detachment underperforms, the flowing-lithium target closes the leg on its own at 0.16 MW m^{-2} [32].

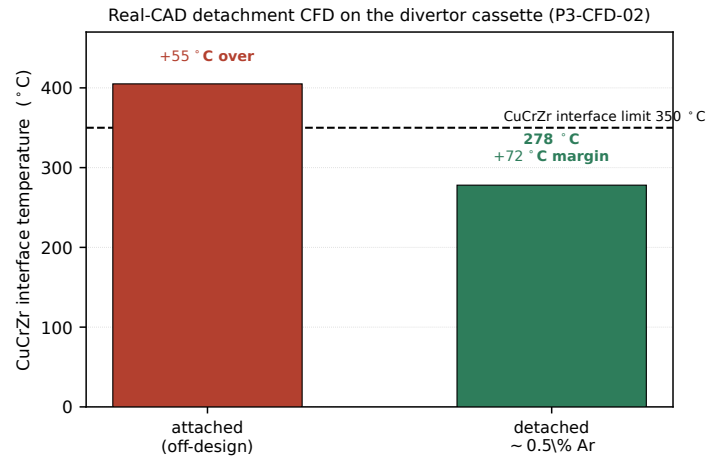


Figure 8. Real-CAD detachment CFD on the divertor cassette (P3-CFD-02). At $\sim 0.5\%$ argon detachment the CuCrZr interface holds 278°C , a 72°C margin below the 350°C limit (dashed); the attached off-design case is over-limit ($+55^\circ\text{C}$). Detachment is confirmed necessary and sufficient.

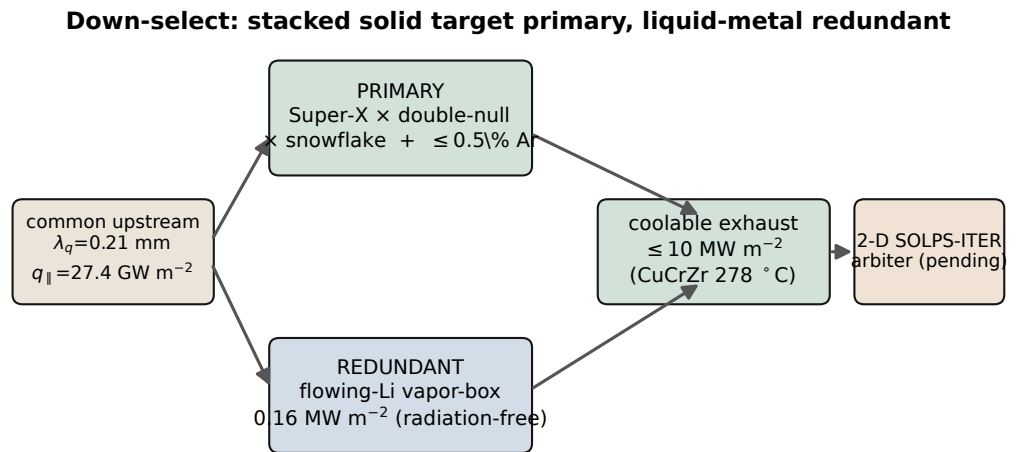


Figure 9. (Schematic.) The selected exhaust portfolio. Against the common upstream boundary, the primary path is the stacked solid target (Super-X $\times$ double-null $\times$ snowflake at $\leq 0.5\%$ Ar) and the redundant path is the flowing-lithium vapor-box; both deliver a coolable exhaust ($\leq 10 \text{ MW m}^{-2}$), and a full 2-D SOLPS-ITER solve is the remaining arbiter.

5.2 Against the published divertor literature

The result sits comfortably inside, and is deliberately more conservative than, the reactor-divertor literature. SPARC faces the same physics: an unmitigated parallel heat flux $>10 \text{ GW m}^{-2}$, closed only by deep radiative dissipation [5]; the Hyperion throat of 27.4 GW m^{-2} is higher still because the compact high-current ST narrows λ_q further (equation 3), which is precisely why geometry, not radiation alone, must carry the first two orders of magnitude. The ITER tungsten divertor is designed around nitrogen/neon seeding to hold its targets [6, 7], and our $\sim 0.8\%$ nitrogen (or $\leq 0.5\%$ argon on the stacked geometry) is in the same regime while radiating a comparable $\sim 90\text{--}94\%$ fraction. The Super-X reduction we report ($8\times$) is consistent with the flux-expansion mechanism analysed by Valanju and Kotschenreuther [11, 12] and demonstrated on MAST-U, where the long-legged Super-X accesses detachment at lower upstream density and with a wider window than a conventional leg [16]. The snowflake contribution — multi-leg flux sharing and an enlarged low-field volume from the quadratic-null field structure [13, 14] — is the same lever explored on NSTX and TCV, here quantified as the step that lifts the Monte-Carlo closure from marginal to robust. The liquid-metal path follows the lithium-vapor-box concept of Goldston and co-workers [18], with the volumetric vapor processes of equation (11) providing a closure mechanism orthogonal to impurity radiation.

5.3 Why the detachment window is broad

A recurring worry for reactor detachment is controllability: a narrow window in the control variable makes the detachment front unstable to hold [16, 17]. Two features widen the Hyperion window. First, the Lengyel cooling function is broadly peaked, so f_{rad} saturates and the margin above 90% dissipation is wide (figure 5). Second, the advanced geometry lengthens the connection length and enlarges the low-field volume (the Super-X and snowflake terms in equations 4 and 9), which is exactly the configuration change shown to enlarge the detachment window [16]. The real-time regulation of this window — radiated-power-fraction control with a radiative-collapse guard — is developed in a companion control study [33] and is not re-derived here.

6 Limitations

One honest gate bounds the claims. The stacked-detachment closure probabilities are from reduced and 1-D edge models plus a Monte-Carlo stack, cross-checked by a 2-D slab scan and a real-CAD CFD; the confirming measurement is a full 2-D SOLPS-ITER solve on the breeder equilibrium, and it is in build. SOLPS-ITER 3.2.1a has been compiled from source and configured on the config-22021 equilibrium (`g_bra9_22021.geqdsk`) meshed with Carre2, but the two-dimensional solve remains grid-topology-blocked (BR-L2-A11/A11b) and has not yet returned a detachment number — nothing is fabricated in its place. That solve refines the seeding budget and the peak-flux ladder to converged 2-D fidelity; it does not gate the selection, which is already backed by three-method detachment agreement (SD1D, UEDGE, Lengyel), a real-CAD CFD holding the copper interface at 278°C , and an independent radiation-free liquid-metal closure.

7 Conclusion

Against a common upstream boundary ($\lambda_q = 0.21 \text{ mm}$, $q_{\parallel} = 27.4 \text{ GW m}^{-2}$), the Hyperion breeder’s divertor exhaust is closed by a stack of levers, not a single geometry. The Super-X cuts the attached target $8\times$ to 171 MW m^{-2} , the double null a further $1.8\times$, and a stacked Super-X $\times$ double-null $\times$ snowflake reaches a coolable detached state at only $\leq 0.5\%$ argon with 95% probability — where the reduced stacks give 52% and 0% . Three independent edge models agree that a sub- 1% seed detaches the leg, a real-CAD CFD confirms the selected detachment holds the copper interface at 278°C , and a flowing-lithium vapor-box provides an independent closure at 0.16 MW m^{-2} . The selected exhaust is the stacked solid target with a liquid-metal redundancy, and the remaining step is a full 2-D edge solve on the breeder equilibrium. The wider lesson is methodological: a reactor exhaust is a down-select over a portfolio of physically independent levers, made auditable by writing each lever as an equation, grounding each number in a frozen register anchor, and ranking the stack by a Monte-Carlo robustness rather than a nominal margin.

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Reproducibility. Every quantitative claim is a frozen anchor in the Kronos Physics De-Risking Register [28] (doi:10.5281/zenodo.22133057): the two-point/Eich SOL boundary (HX-14), the double-null flux split (BR-CX-02), the stacked Super-X Monte-Carlo closure and the flowing-Li vapor-box cap (BR-SX-09), the flowing-Li film thermofluids (HX-16), the UEDGE slab-SOL scan (BR-L2-A21), the SD1D/Hermes-3 1-D solve (HX-14b/15b), and the real-CAD detachment CFD (P3-CFD-02). Each is regenerable from the archived edge decks, the film thermofluid model, and the Monte-Carlo sampler deposited with this paper, all on the frozen config-22021 design point (seed 20260726). This paper is archived at its DOI doi:10.5281/zenodo.22645933; official version: <https://www.kronosfusionenergy.com/publications/a-divertor-exhaust-concept-down-select-for-a-compact-sp.html> and its official page <https://www.kronosfusionenergy.com/publications/a-divertor-exhaust-concept-down-select-for-a-compact-sp.html>; the register is archived at doi:10.5281/zenodo.22133057.

Data availability

The heat-flux ladder, the cross-code detachment inputs, the Lengyel budget, the flowing-Li film points, the Monte-Carlo closure output, and the CFD interface temperatures used for figures 1–8 are archived with the deposit at doi:10.5281/zenodo.22645933, which references the Kronos Physics De-Risking Register doi:10.5281/zenodo.22133057 and the Hyperion breeder design paper [29]. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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